1 Basic Topology

1.1 Exercises

Problem 1.1.1 ▶ Convex Bodies

A convex body is a compact, convex subset $K \subseteq \mathbb{R}^n$ which contains 0 as an interior point and which is symmetric around the origin .

1. Prove that the following formula defines a norm on $\mathbb{R}^n$:

$$N(x) := \inf\left\{\lambda > 0 : \frac{x}{\lambda} \in K\right\}$$

2. Prove that K is homeomorphic to $\mathbb{D}^n$ and that ∂K is homeomorphicc to $\mathbb{S}^{n-1}$.

Note.

1. 若 K 是一个单位球. 则 $N(x) = \|x\|$.
2. 证明 $N(x)$ 是一个 norm, 需要证明:1. 非负正定 2. 绝对一次齐次 3. 三角不等式.
3. 由于 φ 的形式足够简单, 可以通过直接构造逆映射证明双射.

Lemma 1.1.2

X is compact, Y is Hausdorff . If $\exists f : X \to Y$ is a continious bijection, then X, Y are homeomorphic.

Proof. 1. It is obvious that $N(x) \geq 0$. If $N(x) = 0$, then

$$\inf\left\{\lambda > 0 : \frac{x}{\lambda} \in K\right\} = 0$$

which means that for all $\varepsilon > 0$, $\frac{x}{\varepsilon} \in K$ .Since K is compact, $\mathrm{dist}(K, 0) < \infty$. If $x \neq 0$,then $\|x\| > 0$. We take $\varepsilon_0 < \frac{1}{\mathrm{dist}(K,0)\cdot\|x\|}$. Then $\left\|\frac{x}{\varepsilon_0}\right\| > \mathrm{dist}(K, 0)$. But $\frac{x}{\varepsilon_0} \in K$, which is a contradiction.For the second part, let

$x \in \mathbb{R}^n, k \in \mathbb{R} \setminus \{0\}$. For all $\lambda > N(x)$,

$$\frac{x}{\lambda} \in K \implies \frac{kx}{k\lambda} \in K$$

,which shows that $N(kx) \leq kN(x)$. Replace k by $\frac{1}{k}$ and replace x by kx, we have $N(kx) \geq kN(x)$. Thus $N(kx) = kN(x)$. For the triangle inequality , we need to show that

$$\inf\left\{\lambda > 0 : \frac{x_1 + x_2}{\lambda} \in K\right\} \leq \inf\left\{\lambda > 0 : \frac{x_1}{\lambda} \in K\right\} + \inf\left\{\lambda > 0 : \frac{x_2}{\lambda} \in K\right\}$$

Let $\lambda_0 = N(x_1 + x_2)$. Only need to show that

$$\frac{x_1 + x_2}{N(x_1) + N(x_2)} \in K$$

For all $\delta > 0$, we have

$$\frac{x_1}{N(x_1) + \delta} \in K, \quad \frac{x_2}{N(x_2) + \delta} \in K$$

. Since K is convex , we have the convex combination

$$\frac{1}{N(x_1) + N(x_2) + 2\delta}\left((N(x_2) + \delta)\frac{x_2}{N(x_2) + \delta} + (N(x_1) + \delta)\frac{x_1}{N(x_1) + \delta}\right) \in K$$

That is

$$\frac{x_1 + x_2}{N(x_1) + N(x_2) + 2\delta} \in K$$

for all $\delta > 0$. Then

$$N(x_1 + x_2) \leq N(x_1) + N(x_2) + 2\delta, \quad \forall\delta > 0 \implies N(x_1 + x_2) \leq N(x_1) + N(x_2)$$

2. $x \in K^\circ \iff N(x) < 1$, $x \in \operatorname{Ext}(K) \iff N(x) > 1$, $x \in \partial K \iff N(x) = 1$. We define a map $\varphi : \mathbb{R}^n \to \mathbb{R}^n$.

$$\varphi(x) := \begin{cases} N(x)\frac{x}{\|x\|}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

We have

$$\|\varphi(x)\| = N(x)$$

Then φ maps K° on to $(\mathbb{D}^n)^\circ$, and maps ∂K on to $\mathbb{S}^{n-1}$. To show φ is continious, only need to show it is continious at 0. In fact, $\|\lim_{x\to 0}\varphi(x)\| = \lim_{x\to 0}\|\varphi(x)\| = \lim_{x\to 0} N(x) = 0$ since N is a norm. We define $\psi : \mathbb{R}^n \to \mathbb{R}^n$:

$$\psi(x) := \begin{cases} \frac{\|x\|}{N(x)}x, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

Then $\varphi \circ \psi = \psi \circ \varphi = \mathrm{Id}$. The same skills can be used to show ψ is continious at 0 as well. Finally, by the lemma above, X, Y are homeomorphic.

□

Problem 1.1.3

Let X, X' be topological spaces and $x \in X$, $x' \in X'$ be base points. Prove that the wedge sum $X \vee X'$ is homeomorphic to the subspace:

$$(X \times \{x'\}) \cup (\{x\} \times X') \subseteq X \times X'$$

Proof. $X \vee X' = (X \coprod X') / \{x \sim x'\}$. Considering map $\varphi : X \times X' \to X \vee X'$

$$\varphi(x_1, x_2) = \begin{cases} x_1, & x_2 = x' \\ x_2, & x_1 = x \end{cases}$$

$$\varphi(x_1, x_2) = (1 + x_2 - x')(x_1)$$

□

Problem 1.1.4

The torus $\mathbb{T}$ is the quotient of $[0,1]^2$ by the equivalence relation generated by $(0, y) \sim (1, y)$ for all $x, y \in [0,1]$. Prove that $\mathbb{T}$ is homeomorphic to:

1. The product $\mathbb{S}^1 \times \mathbb{S}^1$
2. The quotient of $\mathbb{R}^2$ under the action of the (discrete) group $\mathbb{Z}^2$ by translation.

Problem 1.1.5 ▸ Projective spaces

Let $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. The n-dimensional projectinve space $\mathbb{KP}^n$ is the orbit space of $\Bbbk^{n+1} \setminus \{0\}$ under the action of $\Bbbk^*$ by rescaling.

1. Prove that $\mathbb{RP}^n$ is homeomorphic to the orbit space $\mathbb{S}^n \backslash \{\pm 1\}$, where $\{\pm 1\}$ acts by multiplication on $\mathbb{S}^n \subseteq \mathbb{R}^{n+1}$.
2. Prove that $\mathbb{CP}^n$ is homeomorphic to the orbit space $\mathbb{S}^{2n+1} \setminus \mathbb{S}^1$, where $\mathbb{S}^1 = \{z \in \mathbb{C} : |z| = 1\}$ acts on $\mathbb{S}^{2n+1} \subseteq \mathbb{C}^{n+1}$ by multiplication.
3. Prove that $\mathbb{RP}^1$ is homeomorphic to $\mathbb{S}^1$ and that $\mathbb{CP}^1$ is homeomorphic to $\mathbb{S}^2$.

Problem 1.1.6

1. Let A be a compact subspace of a Hausdorff space X. Prove that X/A is Hausdorff.

Proof.

Lemma 1.1.7

G 作用在 X 上. 假设 G 作用是 proper 的.

$$\begin{aligned} G \times X &\to X \times X \\ (g, x) &\mapsto (g \cdot x, x) \end{aligned}$$

$\forall [x], [y] \in X/A$.

1. $x \notin A, y \notin A$. Let $p : X \to X/A$ is the quotient map. Take a neibourghood of x, y, U_x, U_y, such that $U_x \cap U_y = \varnothing$, $U_x \cap A \neq \varnothing, U_y \cap A \neq \varnothing$.

   Lemma 1.1.8

   设 X 是一个 Hausdorff 空间, K 是 X 的一个紧子集, p 是 X 中一个不属于 K 的点 (即 $p \in X \setminus K$). 那么, 存在两个不相交的开集 U 和 V, 使得 $K \subset U$ 且 $p \in V$.

2. $x \in A, y \notin A$

$\square$